

US Citizen

EDUCATION

University of Maryland — College Park, MD

Expected Graduation: May 2026

Bachelor of Science in Computer Science, Machine Learning Specialization

GPA: 3.80

Relevant Coursework: Object Oriented Programming, Intro to Computer Systems, Algorithms, Discrete Structures, Organization of Programming Languages, Web Application Development, Intro to Data Science

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML, CSS, React, OCaml, C, SQL

Other: Git, Docker, Flutter, Flask, Express, Android, Google Firebase, MongoDB, Linux, Node.js

EXPERIENCE

Technology Intern — Capital One

August

- Participating in the Capital One Technology Minimester, where I was one of 50 selected from 2000 applicants to learn from Capital One engineers through technical projects
- Building a chatbot curated towards prospective applicants by integrating and finetuning ChatGPT's API, gaining hands-on experience in artificial intelligence and API integration.
- Acquiring practical skills in **Java** and **Node.js**, enabling the development of sample server-side applications and enhancing my ability to build scalable and efficient software solutions

Software Engineer Intern — Marriott International

May 2024 - Present

- Developing **Python** pipelines to automate and validate the removal of over 5 million data entries monthly, reducing manual effort by **80%** leading to significant time savings for the data privacy team
- Leveraging **SQL** and **Snowflake** cloud databases to efficiently manipulate and manage large-scale datasets on the cloud, resulting in a **50%** improvement in data processing speed and accuracy
- Creating a powerful application for users to securely and quickly create, edit, and approve records with the processing power to handle over 3 million data entries, contributing to increased user satisfaction

Machine Learning Researcher — UMD Graduate Faculty

May 2024 - Present

- Conducting advanced research in **Natural Language Processing (NLP)** and **Large Language Models (LLMs)** under the guidance of UMD graduate faculty to help increase advancements in NLP techniques
- Fine-tuning LLM models to enhance their performance and adaptability towards explainable hate speech detection, resulting in improved and low computational cost detection systems with reliability in real-world applications
- Executing a **Retrieval-Augmented Generation (RAG)** system designed to identify and report articles related to child abuse, integrating state-of-the-art NLP techniques and methodologies

Lead Developer — Maryland Fellows Program

February 2023 - Present

- Mentoring and managing a team of 4 junior developers and leading the Web Development Team to achieve project milestones and improve code quality, resulting in higher productivity and enhanced code standards
- Utilizing **React (JavaScript)** and **Node.js** to develop, maintain, and deploy code for the program's application, student, and admin portals, enhancing user experience for over 2,000 active users
- Implementing technologies such as **Docker**, **MySQL**, and **AWS EC2** to elevate and optimize the website's performance, resulting in a **40%** increase in load speed and improved scalability for admins and users

Software Engineer — General Dynamics - App Dev Club

September 2023 - December 2023

- Pioneered and embedded a geofencing and notification management system for a **Flutter**-based app, improving location-based service delivery and user engagement
- Engineered a **Java** topic-modeling machine learning system within a **Flask** application, hosted on Google Cloud Services, to provide automated email summaries, reducing manual report generation time by **60%**
- Integrated the machine learning system into the **Flutter** app to enhance email summary features, ensuring seamless functionality and user experience

PROJECTS

NBA Playoff Predictions | nbapredictor.github.io

May 2024

- Developed a model to predict the outcome of NBA games which utilized Beautiful Soup for web scraping and Pandas for data cleansing
- Utilized Scikit-learn to accurately predict the number of playoff games a given team would win by leveraging regular season data to train 3 different machine learning models, with the best (a decision tree) achieving a mean deviation of less than 2 games
- Generated a net profit surpassing \$700 from sports betting over two months, highlighting advanced prediction skills